VANSH ARORA

Full Stack Developer · AI Engineer · 2× Hackathon Winner · Open to Remote

vansharora2310@gmail.com · linkedin.com/in/vansharora01 · github.com/VanshArora01 · vanshcodes01.onrender.com · 7087780200

EDUCATION

B.Tech — Computer Science & Engineering 2024 – 2027 (6th Sem)
PCTE Group of Institutes, Ludhiana, Punjab **CGPA: 7.21 / 10**

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python
Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	Node.js, Express.js, FastAPI, REST APIs, WebSockets, Event-driven Architecture
Database	MongoDB, Mongoose, NoSQL schema design, indexing
AI / ML	ML model integration via REST, FastAPI inference, real-time data pipelines
Tools & Platforms	Git, GitHub, Razorpay API, Vercel, Postman, Linux, production deployment

TRAINING

Full Stack Development Training — Sensation Software Summer 2024
Covered REST API design, MVC architecture, MongoDB integration, and production deployment workflows

PROJECTS

Disaster Management Portal · *1st Place — Desh Bhagat University 2025* *MERN · WebSockets · REST APIs*

- Architected dual-portal system (Admin + User) with Express.js and JWT-based role authentication using socket.io for real-time communication
- Implemented WebSocket-based real-time alert system, eliminating polling and improving responsiveness during high-frequency updates
- Designed event schema for multi-severity disaster classification and built real-time analytics dashboard with role-based access control

EcoExchange AI · *Hackathon Project* *MERN · FastAPI · ML Model*

- Developed REST API for emissions data processing, storing time-series data in MongoDB and integrating FastAPI-based ML inference
- Designed backend to handle concurrent data ingestion and ML inference through separation of services

Climx · *Hackathon Project* *MERN · GIS APIs · Environmental ML*

- Built API aggregation layer enabling efficient region-based environmental simulations using geospatial queries
- Built analytics dashboard to visualize environmental impact simulations and region-based risk insights using real-time API data .

Dev-OS · *Personal Project* · *In Development* *MERN · FastAPI · LLM*

- Designing backend APIs to persist per-project context (tasks, Git state, session history) in MongoDB
- Integrating LLM via FastAPI to generate context-aware next-step suggestions across projects

Client Projects:

• Developed and deployed 3 production-grade web applications (NGO, industrial, SaaS)
• Implemented secure payment workflows using Razorpay API handling real transaction flows
• Built responsive frontends and deployed systems using Vercel CI/CD pipelines.

DATA STRUCTURES & ALGORITHMS

- Actively solving problems on LeetCode in JavaScript; solid in Arrays, Strings, Hash Maps, Two Pointers, Sliding Window
- Building depth in Recursion, Binary Trees, Graph traversal (BFS/DFS); progressing through Dynamic Programming fundamentals

1st Place — DBU Hackathon 2025 · PCTE Hackathon Winner · Ethical Hacking Workshop · Ideathon Certificate